Metacognitive biases in mental health are not explained away by acquiescence and inattention

A reply to: **Sarna, N., Dar, R., & Mazor, M. (2025).** ***Biased and Inattentive Responding Drive Apparent Metacognitive Biases in Mental Health*****.**

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The ability to recruit and study participants remotely via the internet has become increasingly important in computational psychiatry. This field is relatively young and its methodological standards are not yet firmly established. In a recent study, Sarna, Dar and Mazor (2025) present an important example of how biases in responding (inattention and acquiescence) have the potential to alter estimated associations between self-reported mental health symptoms and metacognition.

Metacognition refers to the capacity to reflect on and self-evaluate mental states, and is thought to be biased or distorted in a range of neuropsychiatric conditions (David, 2020; Wells, 2010). Metacognition is often studied by measuring the level of confidence people report in their decisions on a range of tasks. In this context, work from our labs and others have suggested that transdiagnostic symptom dimensions map onto distinct profiles of metacognitive judgments – with people showing greater symptoms of compulsive behaviour and intrusive thought (CIT) exhibiting heightened confidence, and those with greater anxiety and depression (AD) symptoms exhibiting reduced confidence (for a review: Seow, Rouault, et al., 2021)).

The core claim of Sarna et al. is that these relationships can be explained away by surface level questionnaire-filling behaviour associated with a higher prevalence of inattentive responders in online samples. We agree that the impact of response biases and inattention are important to consider in metacognition research, and their paper adds to a growing understanding of data quality issues in crowdsourcing platforms (Burnette et al., 2022; Goodrich et al., 2023), and associated risks to the validity of research findings (Zorowitz et al., 2023). However, in our view Sarna et al. go too far in suggesting that prior associations between confidence and mental health can be explained away by acquiescence and inattention. In our reply, we highlight how their assertions do not hold in newer datasets that use different recruitment methods and have reduced rates of inattention and acquiescence. We also show that their analysis of new experimental data suffers from methodological problems (including low statistical power and omission of key covariates), which when corrected, substantially undermines their key claim and instead indicates that metacognitive biases remain linked to mental health symptoms after controlling for inattention and acquiescence.

Patient and unpaid participant data faithfully replicate bi-directional associations with mental health first seen in crowdsourced samples. Studies have shown that OCD patients, usually tested in-person, have reduced confidence relative to healthy comparison participants (Hoven et al., 2019; Dar et al., 2022). On this basis, Sarna et al. argue that CIT symptoms (of which OCD symptoms are a prominent component) should be negatively associated with confidence – opposite to the result reported in all large-scale online empirical studies to date, which have observed positive associations between CIT and confidence (Benwell et al., 2022; Boldt et al., 2025; Fox et al., 2023; Fox, McDonogh, et al., 2024; Hoven, Luigjes, et al., 2023; Katyal et al., 2025; Mohr et al., 2024; Rouault et al., 2018; Seow et al., 2025; Seow & Gillan, 2020). Sarna et al. propose that inattention in large online studies 'drives' a sign flip in the observed relationship between OCD/CIT symptoms, because inattentive responders are (i) more confident and (ii) use rating scales more randomly. Because random responding produces elevated scores on positively skewed questionnaires (i.e. of rare aspects of mental illness like OCD), this can introduce a spurious correlation. While this is an important illustration of the impact of inattention in uncontrolled settings, here we demonstrate that this pattern is not obtained in well-controlled online data gathered both from patient samples recruited through closed channels and unpaid 'citizen scientist' volunteers.

Fox et al. (2023) report data from the precision in psychiatry (PIP) study, which studied metacognition in a large online sample of treatment-seeking individuals. Participants were only recruited after registering for a course of internet-based cognitive behavioural therapy (iCBT) for anxiety and depression. In addition to exclusions based on task performance, of the N=645 retained for analysis, only 54 (8.3%) participants failed at least 1 attention check of 6 administered during the 4-week longitudinal study. Attention checks were phrased: "*If you are paying attention to these questions, please select x*". Crucially, and in contrast to Sarna et al., the removal of inattentive responders has no effect on the associations between CIT/AD and confidence (Table S1), with inattentive participants marked in blue below (Figure 1). Notably, this treatment-seeking sample had higher levels of psychopathology and therefore reduced skew in the OCD/CIT scales (Figure S1) – markedly reducing the systematic relationship between inattention and symptom severity that Sarna et al., describe. In fact, the difference between attentive and inattentive participants in CIT scores is not

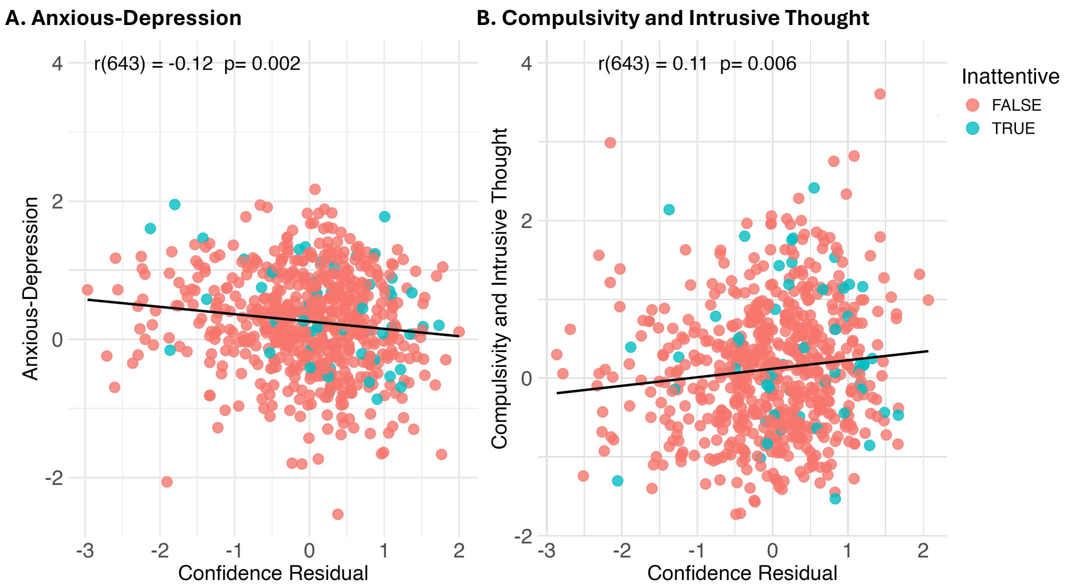

Figure 1. Inattentive responding (in blue) has no impact on the associations between confidence and mental health dimensions in a large clinical sample from Fox et al. (2023). Confidence Residual refers to confidence scores residualised for age and gender and AD/CIT.

significant (Table S2). These data contradict Sarna et al.'s conjecture that positive associations between OCD/CIT and confidence are driven by a combination of inattention and questionnaire skew.

Treatment-seekers in the PIP study were not recruited on crowdsourcing platforms and had lower inattention rates (8% vs 26%) and a smaller bias (3x smaller for the OCD scale; Table S2) than what Sarna et al., report. However, they were still paid to take part. Inattention rates are likely to be even lower in "citizen science" projects, which recruit unpaid, self-motivated volunteers. Importantly, in such a sample, the bidirectional relationships between confidence and CIT/AD can be fully replicated (Fox, McDonogh, et al., 2024). N>3000 citizen scientists downloaded the Neureka smartphone app and completed cognitive tasks and surveys without any financial inducement. Unsurprisingly, this sample of unpaid volunteers had extremely low inattention rates (<1% failed the one infrequency item "*I competed in the 1917 Summer Olympics Games"*). Together, these two datasets refute the argument that inattention accounts for the positive association between CIT and confidence in a variety of large online datasets.

Acquiescence bias is observed in both lab-based and online studies alike. In addition to inattention, Sarna et al. posit acquiescence as a second phenomenon that contributes to a positive association between OCD/CIT and confidence. Acquiescence is the tendency to agree or disagree survey items, regardless of their content, and as Sarna et al. illustrate, this tendency can manifest in higher (/lower) ratings of confidence and mental health questionnaires alike. Although we agree that the phenomenon of acquiescence may well contribute to metacognitive judgments, it is a complex phenomenon that is trait-like (Weijters et al., 2010), and has systematic relationships to various demographics (e.g. age, income and education) (Meisenberg & Williams, 2008; Soto et al., 2008; Ware, 1978). It also predates online crowdsourcing platforms by many decades (Cronbach, 1942) and so it is unclear how this might reconcile the apparent divergence in the metacognition literature. Although Sarna et al. do not state it directly, a key premise of their analysis is that acquiescence is expected to be larger in unpaid citizen scientists compared to lab-based participants (usually paid or credited), rendering online samples more vulnerable to these effects. To evaluate this claim, we estimated the degree of acquiescence in both the PIP study and Neureka datasets described above using the metric developed by Sarna et al., i.e., the degree of inconsistency in the correlation between confidence and standard versus reverse-coded questionnaire items. We found that compared to crowdsourced samples, acquiescence was present, but much reduced, in the PIP study of paid treatment-seekers (Figure S2C). Crucially, it was not observed at all in 3 of the 4 scales in a sample of >800 unpaid Neureka participants (Figure S2D) and there was only a marginal difference between reverse and standard items in the 4th scale. Despite this, the bidirectional relationships between confidence and CIT/AD once again replicate (Table S3), contradicting the claim that acquiescence alone drives these effects. Moreover, given the near-absence of acquiescence effects in the Neureka dataset, the claim that acquiescence is larger in unpaid citizen scientists compared to lab-based participants also does not appear credible.

The conclusions about the broader literature rests on an analysis that is seriously underpowered, does not measure AD and CIT, and omits important covariates.

Sarna et al.'s finding that the association between obsessive-compulsive symptom scores (OCI-R scale; Foa et al., 2002) and confidence is non-significant when they control for inattention and acquiescence is difficult to interpret for several reasons. First, it is severely underpowered at N=190 (all) and N=144 (attentive only). In studies using the same perceptual decision-making task, estimates of the Ns required to achieve 80% power for an association between CIT and confidence are N=454 (Rouault et al., 2018), N=542 (Fox et al., 2023) and N=421 (Benwell et al., 2022). Second, Sarna et al. investigate OCI-R and depression scores but seek to generalise their findings to the published literature on CIT and AD respectively. Meta-analytic evidence across 4 datasets comprising N=2,572 participants suggests the association between confidence and CIT/AD is approximately twice the size of that with the OCD/depression scales (Fox, Teckentrup, et al., 2024). Indeed, across two experiments in Rouault et al., (2018) and two experiments in Benwell et al., (2022) there were no significant associations with obsessive-compulsive scores and confidence at all. Beyond statistical power, this highlights how the premise of their study is based on false equivalences between a depression scale (SDS: Zung, 1965) and the AD dimension, and between obsessive compulsive symptoms (with implications to the disorder of OCD) and CIT– an issue we return to later.

Finally, Sarna et al.'s exploratory analysis does not control for other co-occurring clinical symptoms and demographics (age and sex/gender). This is in contrast to the approach taken by all of the large online studies they seek to generalise their findings to. These variables impact the observed associations as much, if not more, than inattention or acquiescence. For example, in their data the positive association between OCD and depression symptoms is larger (r=.39) than the association between OCI-R and their acquiescence measure (r=.27), but these co-occurring symptoms are not accounted for within the same analysis. As Sarna et al. demonstrate for gender, these variables have systematic relationships with inattention, confidence, and mental health. To illustrate the importance of including these well-established confounds, we added these covariates in a re-analysis of their data and recovered the often-reported significant negative association between depression and confidence that Sarna et al. fail to observe. Importantly, we also find that removing inattentive responders does not in fact reduce the associations between confidence and either the depression or OCD symptoms, in contrast to what they report (Table 1), with both OCD symptoms (*p*=.001) and depression (*p*=.002) remaining significantly and bidirectionally linked to confidence. The association between confidence and OCD, when controlling for depressive scores, age and gender, also remains significant even after correcting for acquiescence (*p*=.009) (Table 1)[1].

		Full sample (N=188)		Attentive Only (N=142)		Attentive Only + Acq. Corr. (N=142)	
		Est. (SE)	**p-value**	**Est. (SE)**	**p-value**	**Est. (SE)**	**p-value**
Model 1	Depression	-0.02 (0.01)	0.077	-0.02 (0.01)	0.075	-0.02 (0.01)	0.089
Model 2	OCD	0.05 (0.01)	<.001***	0.03 (0.01)	0.069	0.02 (0.01)	0.193
Model 3	Depression	-0.04 (0.01)	0.002**	-0.05 (0.02)	0.002**	-0.04 (0.02)	0.006**
	OCD	0.06 (0.01)	<.001***	0.05 (0.02)	0.001**	0.04 (0.02)	0.009**
	Age	-0.02 (0.01)	0.195	-0.02 (0.01)	0.153	-0.02 (0.01)	0.126
	Sex	0.06 (0.02)	0.0097**	0.05 (0.03)	0.041*	0.06 (0.03)	0.042*

[1] In the course of our re-analysis, we discovered two errors in the analysis script of Sarna et al.. The first affects the validity of their corrected depression values by assigning them to the wrong participants. The second affects the correction for acquiescence, which was carried out prior to rather than after excluding inattentive responders. Both of these errors were corrected prior to computing the estimates in Table 1.

Table 1. Re-analysis of Sarna et al. (2025). Models 1 and 2 are used by Sarna et al. (2025). Model 3 includes covariates for co-occurring mental health symptoms and demographics (age and sex) variables. "Acq. Corr." refers to acquiescence correction of clinical scores using the mean neutral rating. Two participants without sex data are removed.

Skew is a feature, not a bug, of mental health data. Sarna et al. raise an important point about how different sources of variance can impact factor analysis structures. Factor solutions are based on correlations between questionnaire items and, as a result, are influenced by various characteristics of items beyond the constructs they purport to measure. Sarna et al. discuss reverse coding (Weijters et al., 2013; X. Zhang et al., 2016) and skew as two such sources of variance, but solutions are likely also influenced by temporal window (e.g. 1 week, 1 month), length of Likert scales, anchors, the number of items measuring each construct, and much more (Fabrigar et al., 1999). The computational factor modelling approach (Gillan et al., 2016; Wise et al., 2023) has thus far not addressed the contribution of these sources of variance to factor solutions and we agree that this is an important next step in this research endeavour. This will involve creating questionnaire formats that are balanced and fit for cross-scale analysis.

That said, the assertions made by Sarna et al. regarding the validity of the AD/CIT/SW factor structure (i.e. that it is mostly driven by skew and reverse coding) are *post hoc* and not supported by data. As Sarna et al. acknowledge, skew is not a property of a survey item that is assigned at random; there are common (anxious-depression) and rare (compulsivity) mental health problems. Therefore, the existence of an association between skew and weights is expected. Importantly, the identified 3-factor structure of AD/CIT/SW replicates across a variety of samples, even those with different patterns of skew. For example, the weights produced from a Mechanical Turk sample (Gillan et al., 2016) (N=1413) are highly correlated with individuals about to initiate CBT in the PIP study (Fox et al., 2023) (N=649) (r=.81-.96; Figure S3A). As one might expect, the correlations are higher still between two general population samples, one paid crowdsourced and the other unpaid citizen scientists from Neureka (N=875) (r=.93-.96; Figure S3B). Moreover, associations between OCD and schizotypy also emerge from analyses of distinct questionnaire sets in various general population samples (not crowdsourced) (Chmielewski & Watson, 2008; Sica et al., 2024). The CIT factor has also been externally validated in several studies that link individual differences to task behaviour (e.g., choices) rather than self-reported confidence, eliminating any potential confound of acquiescence bias affecting rating scales. For instance, CIT shows specific relationships with impairments in model-based planning (Gillan et al., 2016), a finding also observed in patients (Voon et al., 2015). This has been replicated in several studies using diverse sampling methods, including in-person participants (Seow, Benoit, et al., 2021), unpaid citizen scientists (Donegan et al., 2023), and people about to start iCBT (Donegan et al., 2024).

How can we reconcile findings from online general population and patient samples?

Sarna et al. hypothesise that CIT symptoms in the general population (of which OCD symptoms are a prominent component) should be negatively (rather than positively) associated with confidence, in line with observations in OCD patients. However, in all of the data discussed here or collected by Sarna et al., we identified no case where the association between OCD symptoms and confidence was negative. Therefore, to make the case that a negative association does exist in non-clinical samples, if tested in-person, Sarna et al. (preprint version 1) cite 4 studies that purport to have shown a negative relationship between OCD symptoms and confidence. The first two studies are cited in error (Seow & Gillan, 2020; Hoven, Rouault, et al., 2023); the non-clinical samples in these papers were both gathered online, not in-lab, and report positive, not negative, relationships between CIT and confidence. The other two studies concern a very specific false biofeedback task with no controls for objective performance, anxious-depression symptoms or demographics (Lazarov et al., 2012; Z. Zhang et al., 2017). Given the abundance of evidence to the contrary from well-controlled studies for either positive or null (and not negative) associations between CIT and confidence, we suggest that the reconciliation needed in the literature is not at the juncture of online versus in-lab studies, but between having a diagnosis of OCD and scoring high on a dimension of CIT.

In addition to OCD symptoms, CIT has contributions from questionnaires assessing schizotypy, alcohol use, impulsivity and eating disorder symptoms (Gillan et al., 2016) – which confer risk for other diagnoses that typically exclude people from OCD research. Distinct from having high levels of CIT, OCD is a syndrome that is fundamentally linked to anxiety (Foa et al., 1999; Ruscio et al., 2010), so much so it was classified as an anxiety disorder until the most recent edition of the DSM (American Psychiatric Association, 2000). As anxiety and depression are both linked to reduced confidence, it is reasonable to expect that this particular constellation of symptoms might manifest as a disorder of doubt (Dar et al., 2000; Samuels et al., 2017). Beyond differences in symptom profile and co-morbidity, there are substantial differences in how clinicians and patients rate symptoms (Banker et al., 2024) and important distinctions between diagnosed patients and individuals who score high on psychopathology questionnaires but have not sought treatment (e.g. greater distress, less insight: García-Soriano et al., 2014; Montoya et al., 1995). Developing a greater understanding of how transdiagnostic dimensions manifest in clinical disorders is essential for computational psychiatry to translate basic findings into clinical applications. However, this does not mean we should expect one-to-one mappings; the shortcomings of diagnosis-based approaches underpin a motivation for undertaking transdiagnostic research in the first place (Hyman, 2010; Insel et al., 2010). Recruiting large samples of patients to parse individual differences in a comparable way to online studies is an important next step if the field is to realise its translational potential.

Moving forward. Whether samples are gathered in person or online, we cannot measure latent constructs in psychology without error or bias. Appropriately powered samples that can estimate the impact of these potential confounds and foster a culture of reproducibility are crucial and we argue that online methods have made considerable advances over smaller in-person studies in this regard. Implicit measures of metacognition may also help understand the links with psychopathology and add context for self-report data, by reducing the acquiescence biases that Sarna et al. suggest affect explicit reports. For instance, measuring uncertainty-guided checking tendencies (Baptista et al., 2021), reminder setting (Boldt et al., 2025) or information seeking (Schulz et al., 2020) can provide evidence that is independent of acquiescence biases or anchor points which affect use of scales. However, implicit measures can also bring new sources of biases, and tap into different psychological constructs such as the extent to which a person with a given trait acts rationally to resolve it (e.g. (Baptista et al., 2021; Boldt et al., 2025; Mohr et al., 2024)).

We wish to end on an optimistic note. Despite our disagreements regarding some of their conclusions, we robustly applaud the work of Sarna, Dar and Mazor (2025). We hope their paper, and this reply, will invigorate discussion, and encourage greater attention to the importance of data quality in online studies. When quantifying metacognition, their work reinforces the importance of considering the statistical distribution of symptoms in a sample, and accounting for the many relevant sources of variance (including response tendencies, and as many as possible clinical and demographic factors). Exploring and documenting these various factors will deepen our understanding of both metacognition and psychopathology. With this, we must recognise that even in datasets with limited evidence of bias, we cannot ever be sure it has been eliminated – some uncertainty always remains.

That said, there is a fine line between explaining a phenomenon, and explaining *away* that phenomenon. Understanding where that line can be drawn as computational psychiatry matures will only be a good thing for science.

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Author contributions. Claire Gillan and Tricia Seow analysed the data. Claire Gillan, Marion Rouault, Steve Fleming and Tricia Seow wrote the manuscript.

Code/Data availability: Code and data pertaining to this paper are available here: https://osf.io/df4wv

Supporting Analyses

	Full sample		Inattentive Excluded	
	Estimate (SE)	**p-value**	**Estimate (SE)**	**p-value**
Intercept	3.85 (0.04)	< 2e-16 ***	3.82 (0.04)	< 2e-16 ***
AD	-0.11 (0.03)	0.001 **	-0.10 (0.03)	0.005 **
CIT	0.10 (0.03)	0.004 **	0.10 (0.04)	0.005 **
Age	0.05 (0.03)	0.161	0.05 (0.03)	0.188
Sex	0.10 (0.04)	0.013 *	0.09 (0.04)	0.026 *
Education	-0.10 (0.03)	0.004 **	-0.10 (0.03)	0.006 **

Table S1. Statistical results for the Precision in Psychiatry (PIP) study (Fox et al., 2023) data, with and without removing inattentive responders.

	PIP Study			***Sarna et al., 2025***		
	Attentive N=591; 92%	**Inattentive N=54; 8%**		**Attentive N=145;74%**	**Inattentive N=50; 26%**	
	Mean(SD)	Mean(SD)	Cohen's d	Mean(SD)	Mean(SD)	Cohen's d
CIT	0.11(0.82)	0.29(0.89)	0.22	-	-	-
OCI-R	22.7(12.0)	26.5(13.8)	0.31	17.6 (12.8)	30.2 (15.4)	0.94
Confidence	3.76(0.84)	4.10(0.85)	0.40	0.54 (0.17)	0.65 (0.17)	0.62

Table S2. Comparison of the effect size (Cohen's d) of the difference between attentive vs inattentive responders on CIT, OCI-R and Confidence ratings between PIP study and Sarna et al., 2025.

For OCI-R, the impact of inattention on scores is estimated to be 3x larger in Sarna et al., 2025 than the PIP study. The rate of inattention is also 3x larger. Sarna et al., do not measure CIT, but it appears to be less affected by inattention than the OCI-R in the PIP study, with the group difference being non-significant (p=.14).

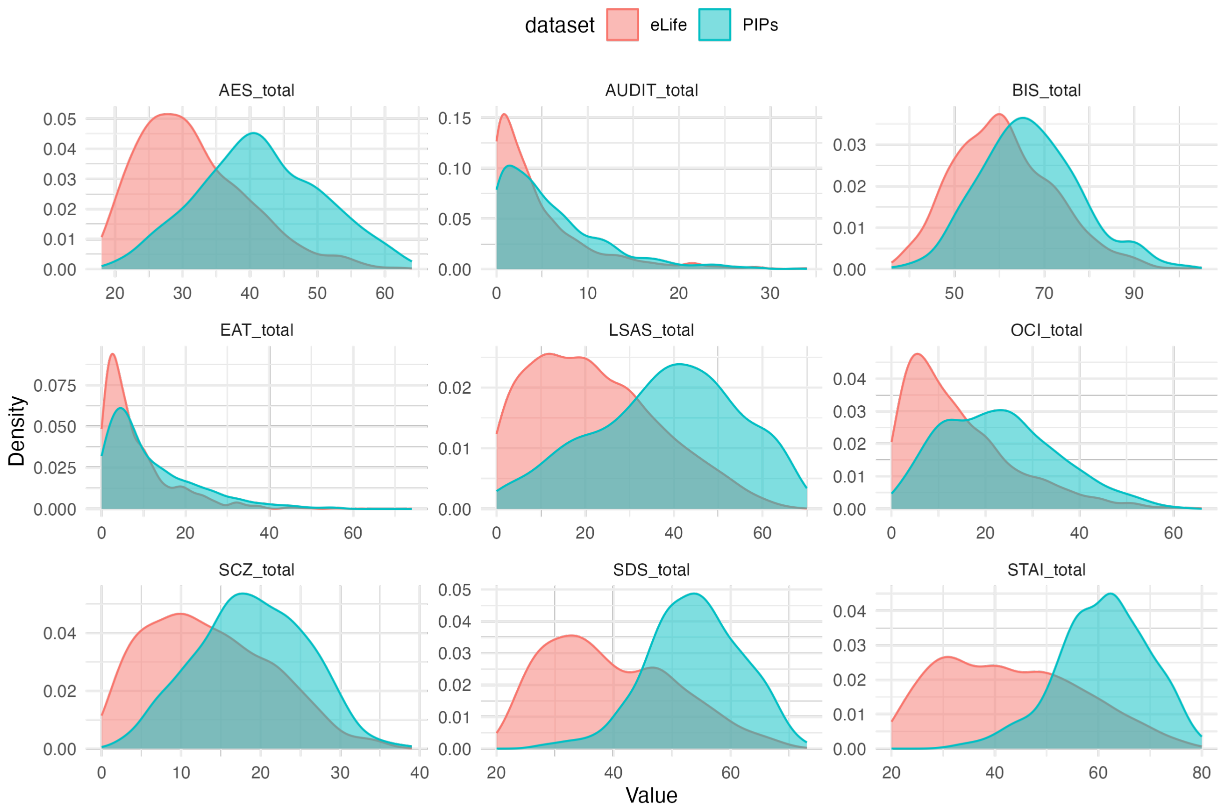

Figure S1. As expected, we observe different patterns of questionnaire skew in the PIP study (clinical sample of treatment-seeking individuals) (Fox et al., 2023) versus Mechanical Turk data (general population sample; eLife) (Gillan et al., 2016).

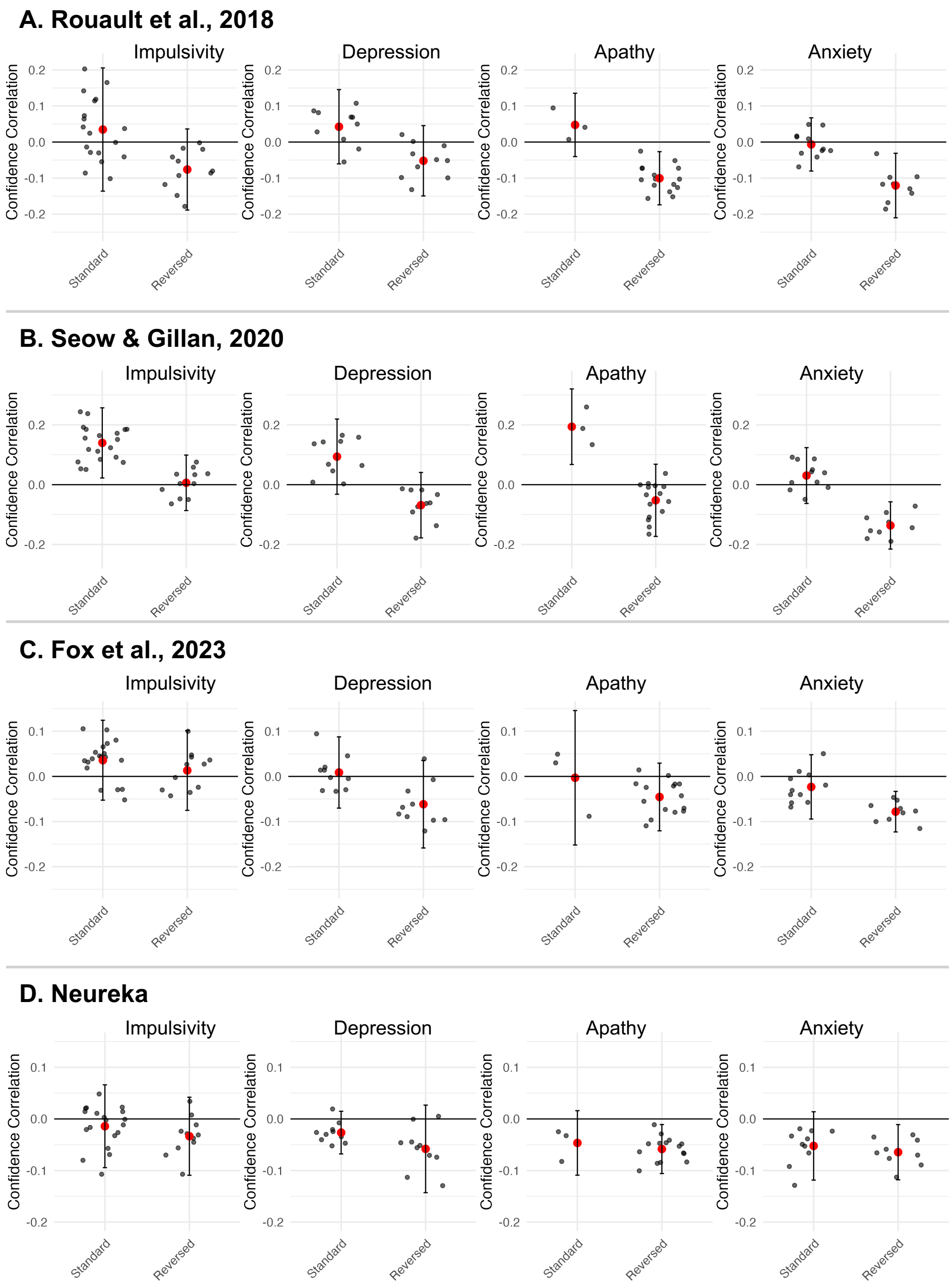

Figure S2. Comparison of confidence correlation between standard and reverse coded items in (A) Rouault et al. (2018), (B) Seow and Gillan (2020), (C) Fox et al. (2023) and (D) Neureka for four questionnaires measuring impulsivity, depression, apathy and anxiety symptoms. *Each black dot represents a questionnaire item. There is no statistical evidence for reverse-coding effect on confidence correlation in the Citizen Scientist 'Neureka' sample (N=875) in ¾ scales [BIS: t(22.08)=1.32, p=.20; AES: t((2.48)=0.62, p=.59; STAI: t(18)=0.91, p=.37] and only a marginal effect for the SDS depression scale [SDS: t(13.77)=2.07, p=.054]. Note that these analyses are not controlled for age and sex (as per Sarna et al., 2025). For a meaningful interpretation of direction and magnitude of associations, it is important to control for these confounds.*

**The published Neureka study (Fox et al., 2024) did not use the full questionnaire set, so we show here a related unpublished Neureka dataset which included all surveys. In this sample, like the published study of >3000, we once again replicate the bidirectional associations between confidence and mental health (Table S3, below).*

	Neureka *'Remember This'*	
	Estimate (SE)	**p-value**
Intercept	0.00 (0.04)	0.939
AD	-0.07 (0.04)	0.052*
CIT	0.08 (0.04)	0.032*
Accuracy	0.14 (0.04)	7.73e-05 ***
Age	0.10 (0.04)	0.007**
Male Gender	-0.01 (0.08)	0.884
Education	0.03 (0.03)	0.371

Table S3. Statistical model for Citizen Scientist sample from the Neureka app (N=875) replicating the bidirectional associations between AD, CIT and confidence.

Remember This is a semantic knowledge quiz concerning Alzheimer's Disease knowledge where each True/False factual question is followed by a confidence assessment. As there is no staircase to control for objective performance at the quiz, accuracy must be controlled for directly in the model.

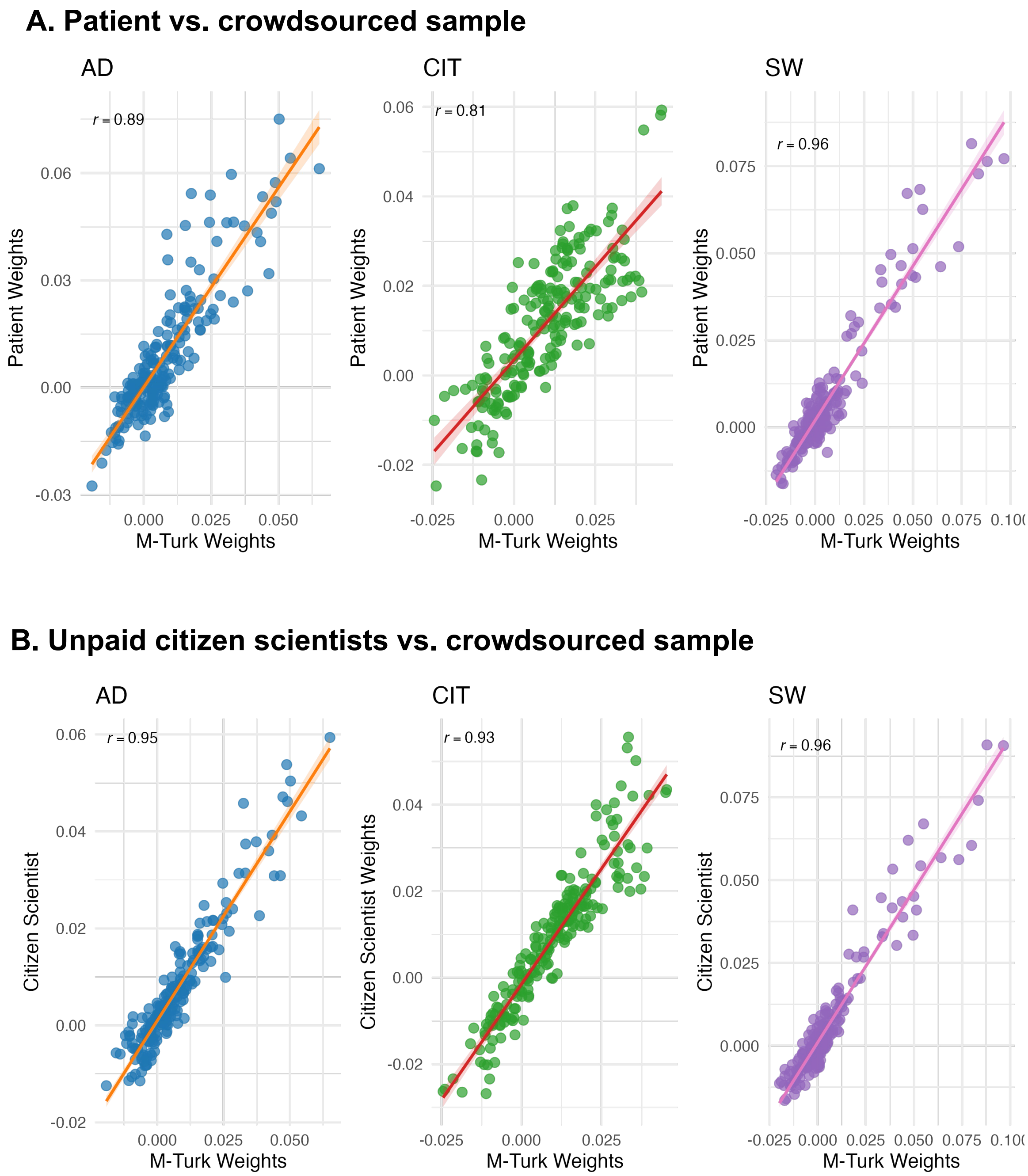

Figure S3. Factor weights are highly correlated across different samples.

(A) Comparison of weights from factor analysis in a Mechanical Turk sample (x-axes) (Gillan et al., 2016) (N=1413) and treatment seeking individuals in the PIP study (y-axes) (Fox et al., 2023) (N=649). (B) Comparison of weights from the same Mechanical Turk sample (x-axes) (Gillan et al., 2016) (N=1413) and an unpaid citizen scientist sample (Neureka, unpublished N=875). AD= Anxious-Depression, CIT= "Compulsivity and Intrusive Thought" and SW= "Social withdrawal".

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